From Hidden Writing to Provable Security

Encryption Function with Key K

Plaintext Message

$$D_K(E_K(M)) = M$$

Decryption Function with Key K

Provable Correctness & Security

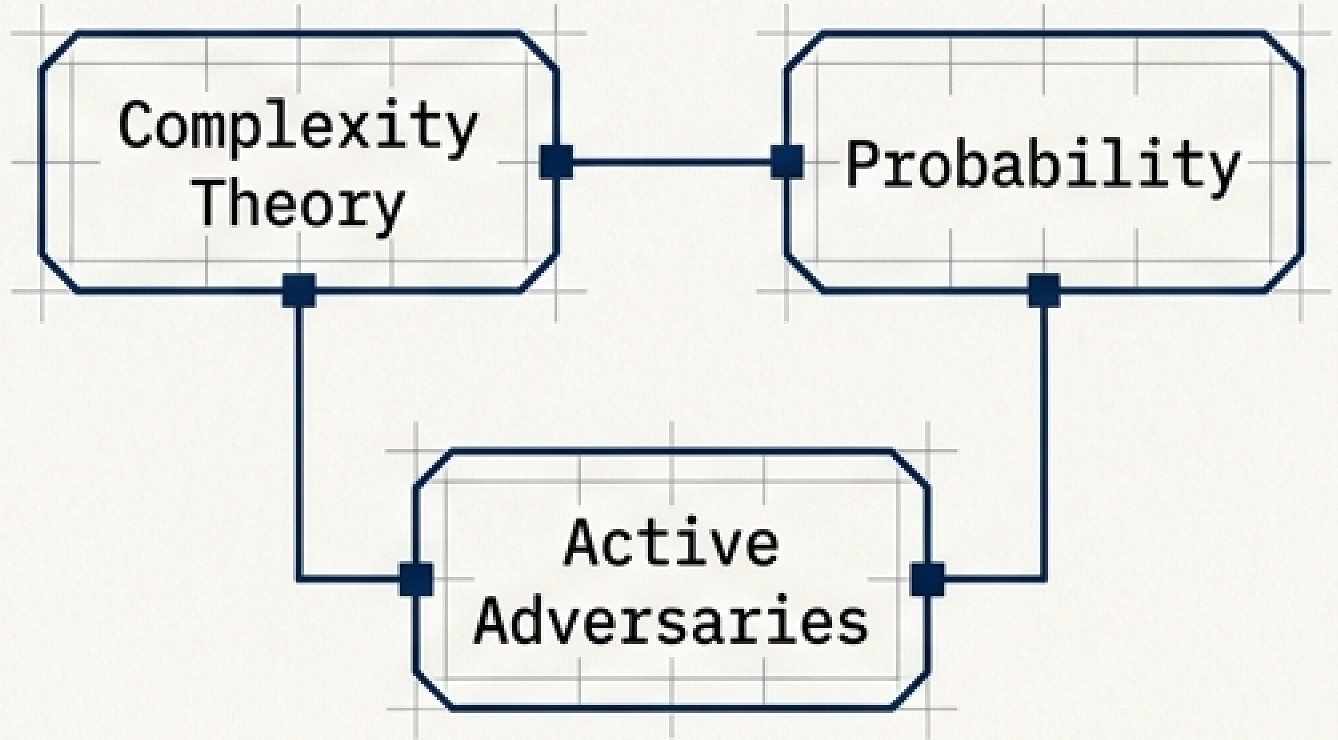
A visual primer on the **modern science** of cryptology, adversary modeling, and the **mathematical architectures** of **trust.**

The fundamental definition has changed

Classical Cryptography

Historically defined as ‘hidden writing.’ The sole goal was concealing information from human interception through heuristic transformations.

Modern Cryptography



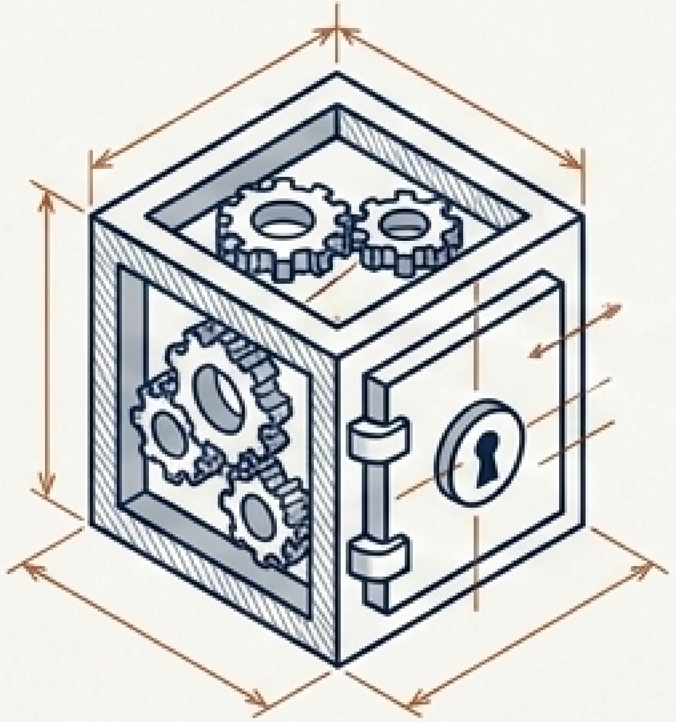
A scientific, mathematical discipline. It provides a rigorously defined collection of security services to protect systems against explicitly defined classes of attacks.

The anatomy of cryptology encompasses builders and breakers

A cryptographic construction is never considered secure simply because it appears complicated. It must survive complementary, rigorous analysis.

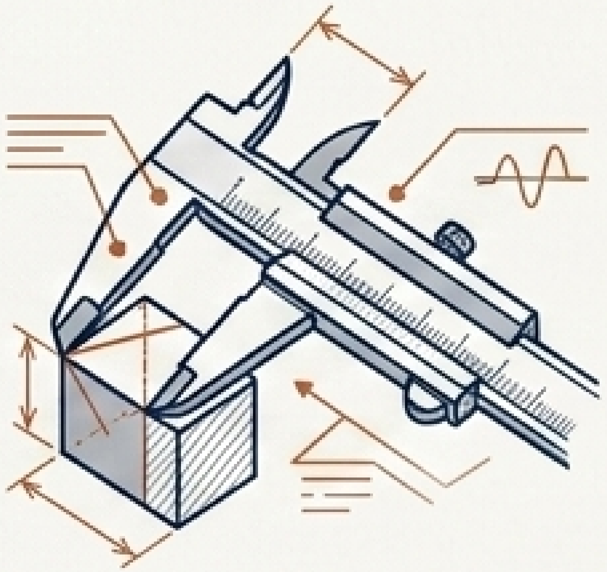
Cryptology

- **Cryptography**
- **Cryptanalysis**

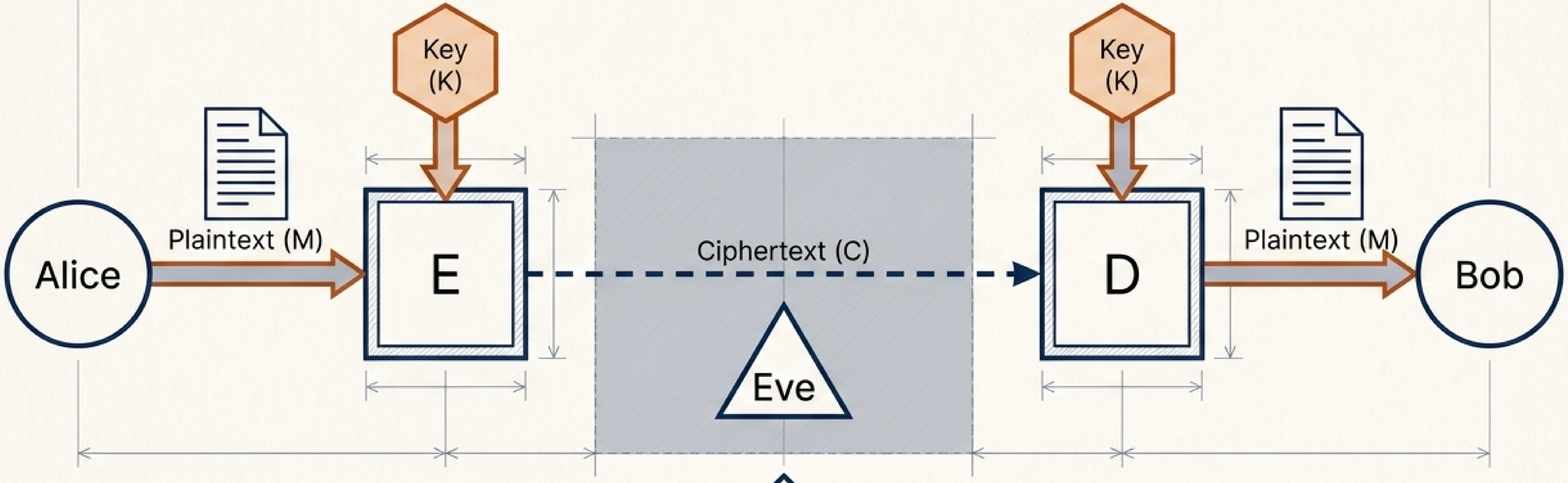


Cryptography: Designing mathematical mechanisms and protocols that guarantee specific security properties.

Cryptanalysis: Analyzing mechanisms to bypass them, extract secrets, or prove they fail their intended security goals.



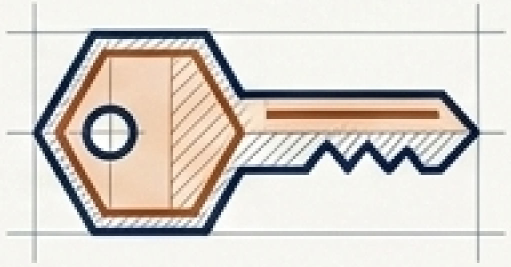
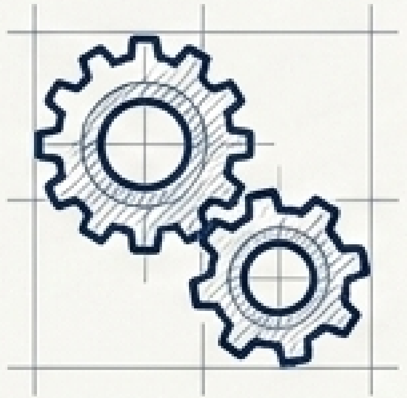
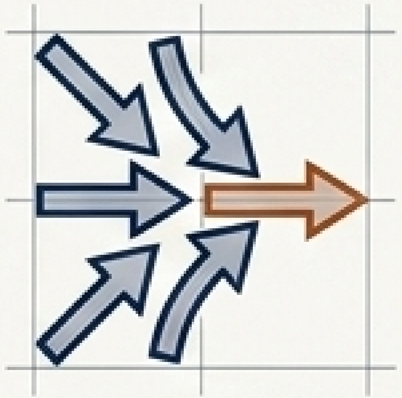
The core cryptographic engine



Strict Mathematical Requirement

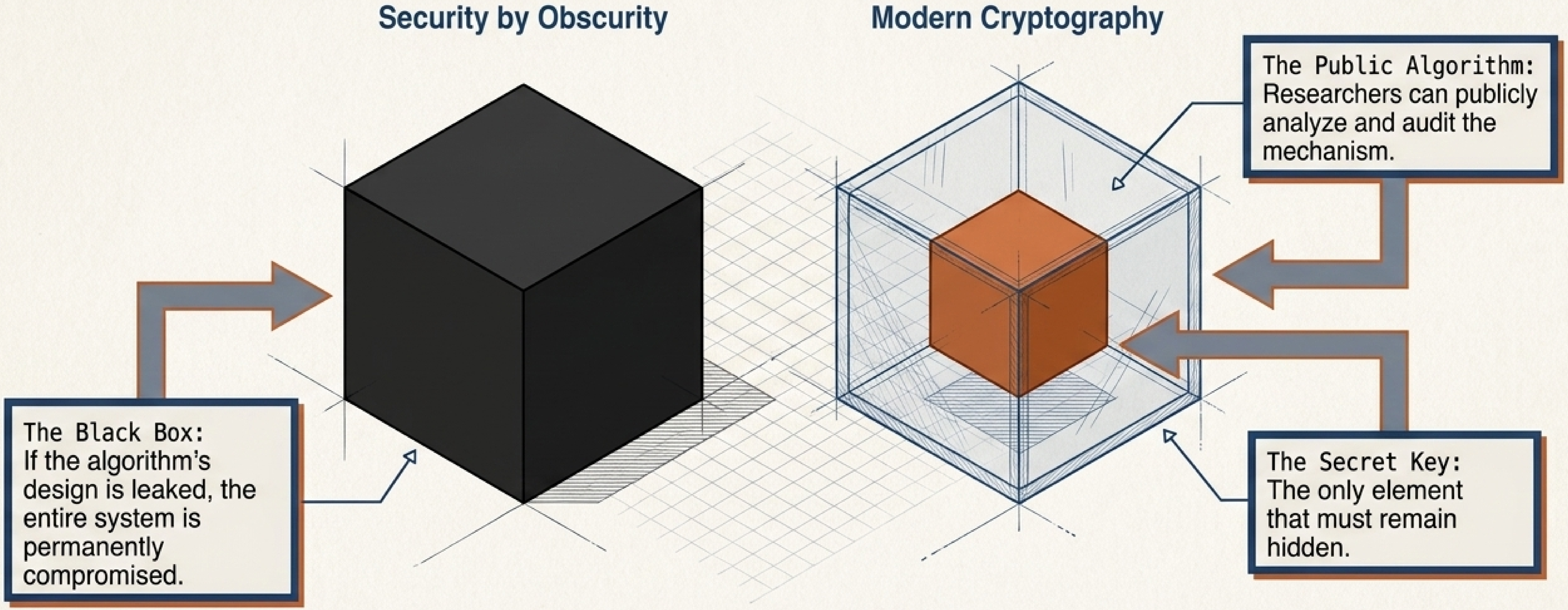
$$C = E_K(M)$$
$$M = D_K(C)$$
$$D_K(E_K(M)) = M$$

Distinguishing encryption from adjacent data transformations

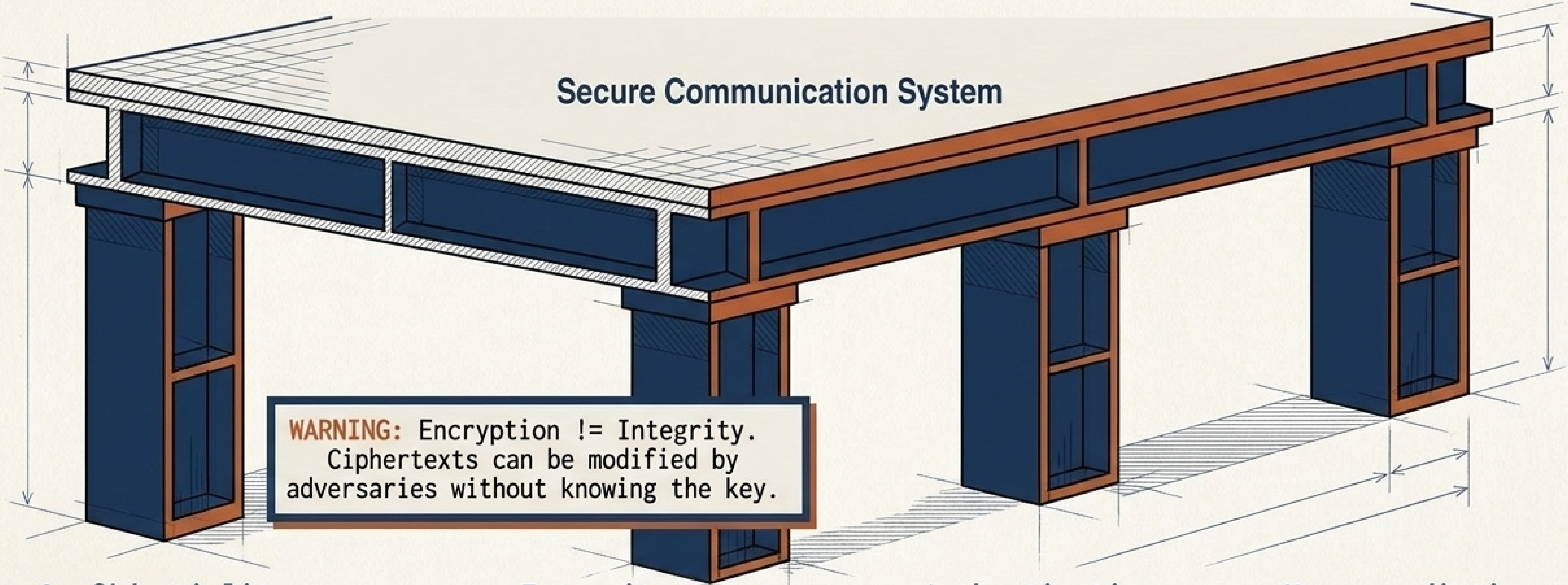
	Encryption	Encoding	Compression	Hashing
				
Goal	Confidentiality	Data Format Translation	Size Reduction	Integrity / Authentication
Mechanism & Reversibility	Reversible only with a cryptographic key.	Fully reversible by anyone. (e.g., Base64)	Fully reversible by anyone. Uses fewer bits.	Irreversible by design. Maps arbitrary inputs to fixed digests.

Kerckhoffs's Principle separates algorithmic secrecy from key secrecy

Security relies entirely on the secrecy of the key, never on the secrecy of the algorithm.



Modern systems require four load-bearing pillars of security



Confidentiality
Preventing unauthorized parties from extracting meaningful information (e.g., Encryption).

Integrity
Detecting unauthorized modification of data (e.g., MACs).

Authentication
Verifying the origin of a message or the identity of an entity.

Non-repudiation
Providing cryptographic evidence associating an action with an entity.

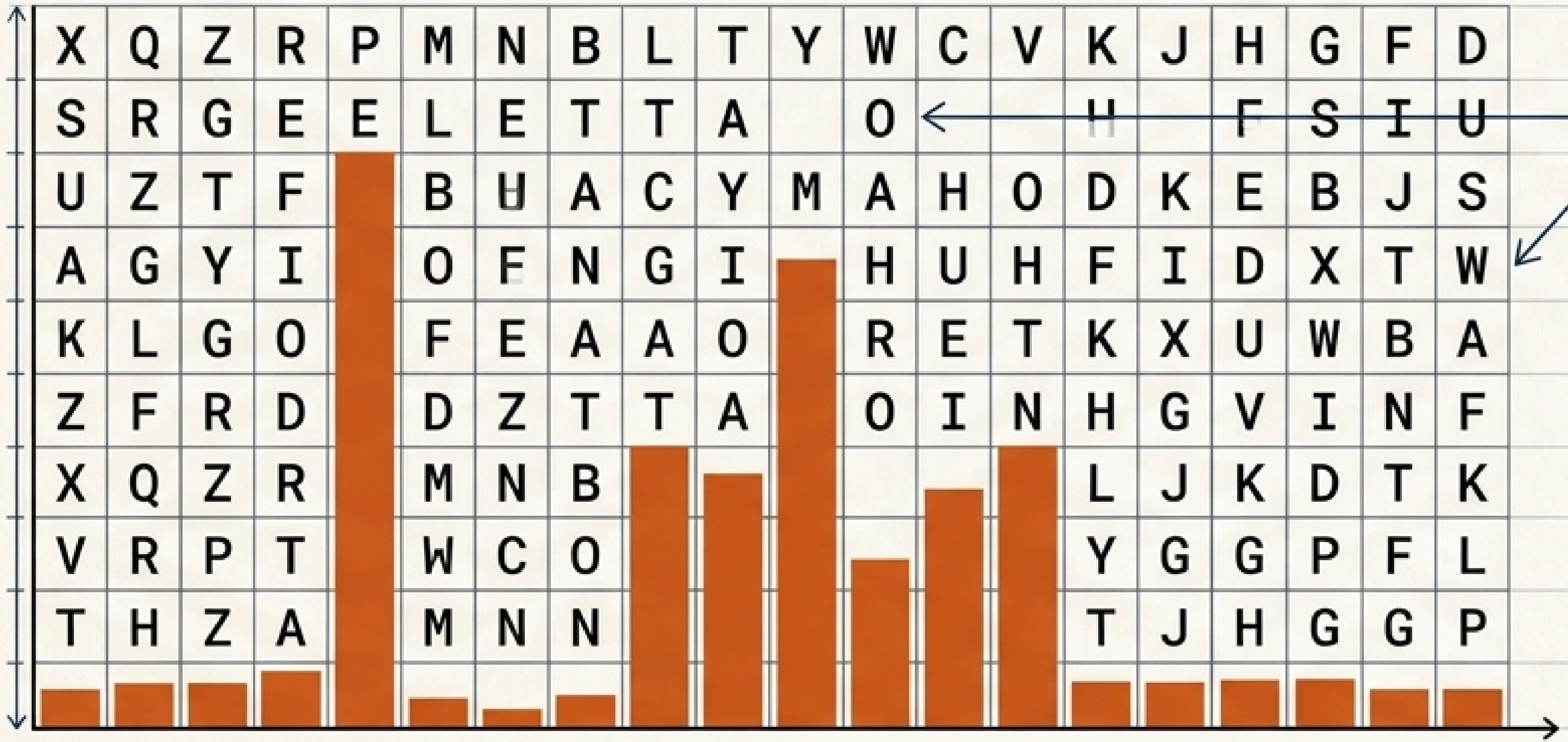
Statistical structure defeats classical obscurity

Classical Mechanisms

Classical cryptography relied on rearranging or substituting human symbols. Vulnerable to exhaustive search.

Transposition: $C = m_{\pi(n)}$

Substitution (Caesar): $E_k(x) = (x+k) \bmod 26$



Frequency Analysis

Monoalphabetic substitutions preserve the statistical structure of natural languages.

Adversaries infer the mapping without needing the key.

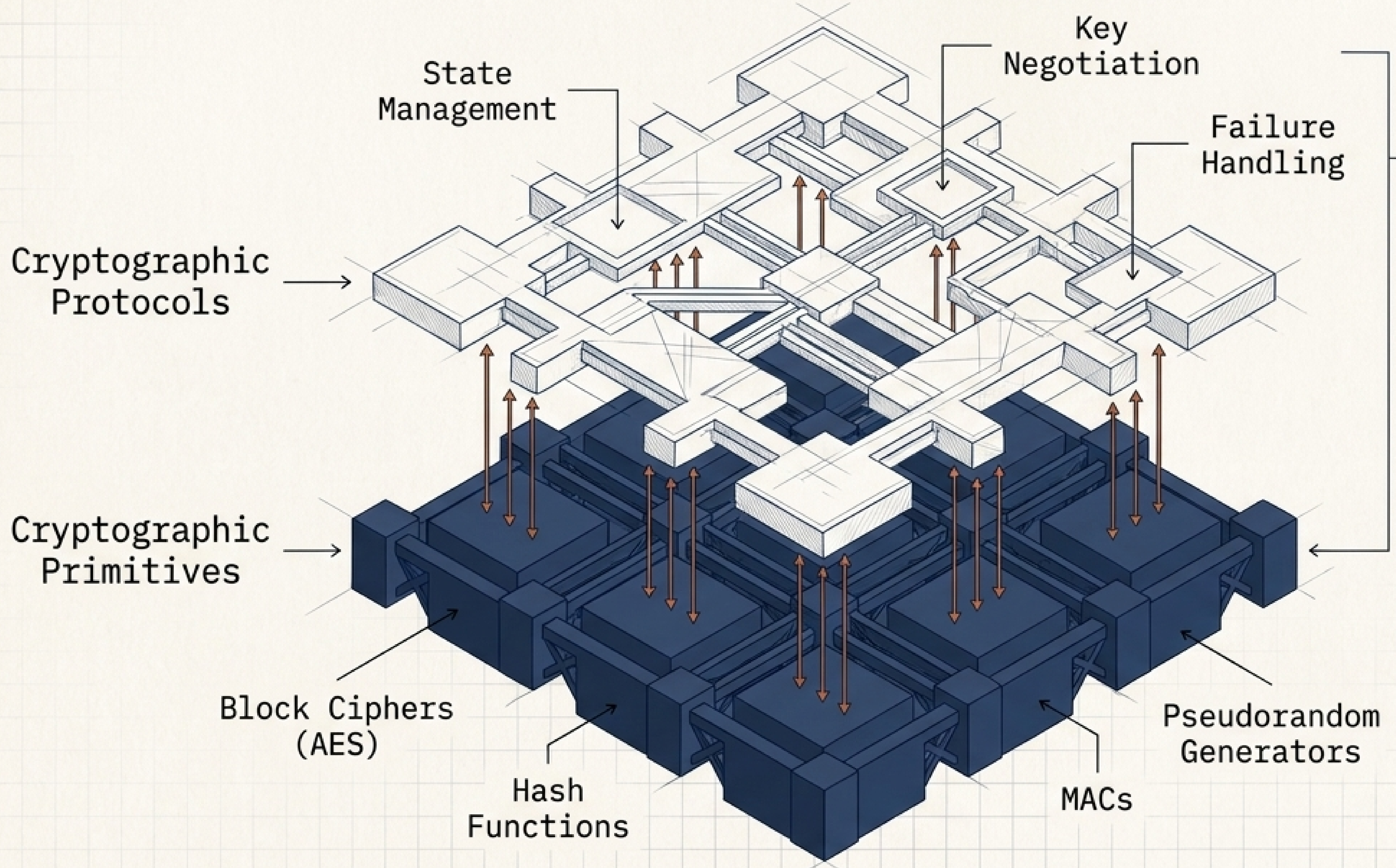
Apparent obscurity is not a measure of security.

Algorithms are evaluated against explicit adversary models

Modern cryptography assumes the attacker is on the network and can capture, replay, modify, and interact with messages.

Adversary Model	What the Attacker Has / Can Do	Adversary Goal
Known-Ciphertext	Possesses only captured ciphertexts.	Extract underlying plaintext without the key.
Known-Plaintext	Possesses pairs of (M, C) encrypted under the same key.	Infer the secret key or read other messages.
Chosen-Plaintext (IND-CPA)	Can submit chosen messages to be encrypted and observe outputs.	Distinguish encryptions of chosen messages.
Chosen-Ciphertext (IND-CCA)	Can submit ciphertexts to a decryption mechanism and observe results.	Break the system through active, interactive manipulation.

Primitives are the mathematical building blocks of protocols

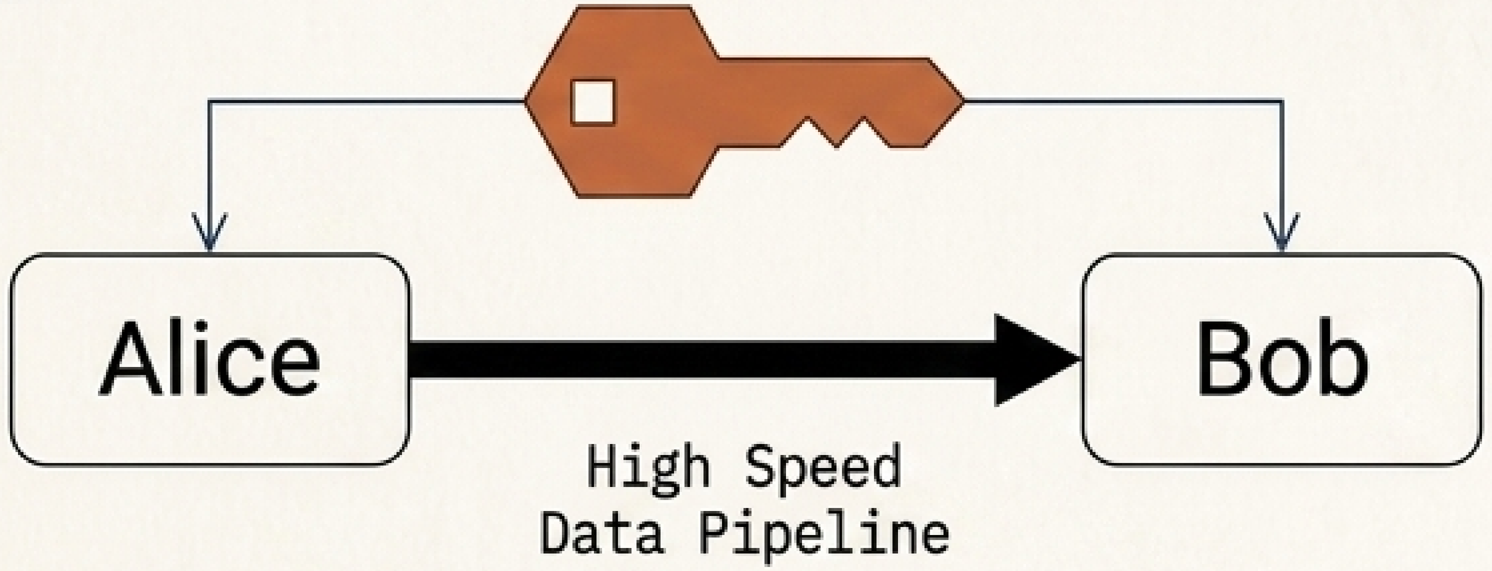


A protocol can be critically vulnerable even if every underlying mathematical primitive is perfectly secure.

Security must be analyzed at the structural protocol level.

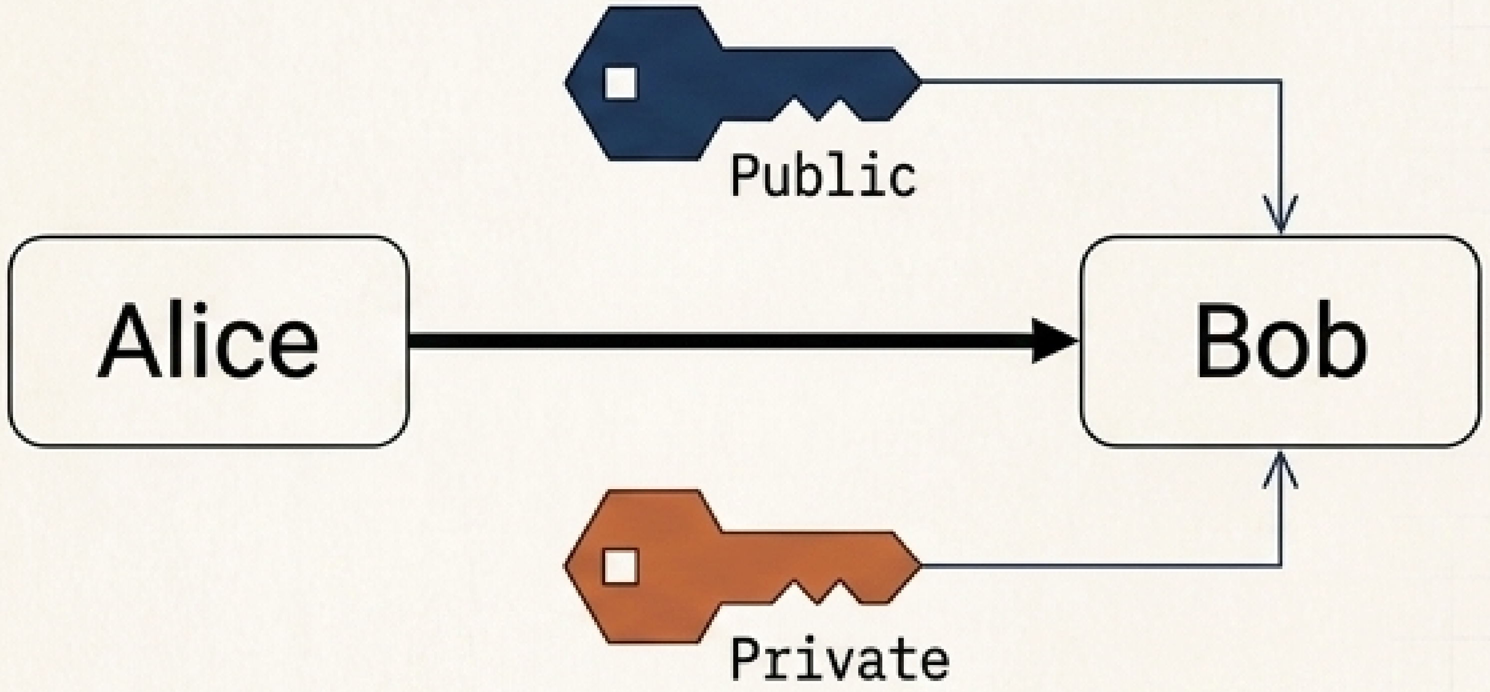
Symmetric efficiency versus asymmetric capability

Symmetric Cryptography



Communicating parties share a single secret key. Highly efficient, essential for protecting large volumes of data (e.g., AES).

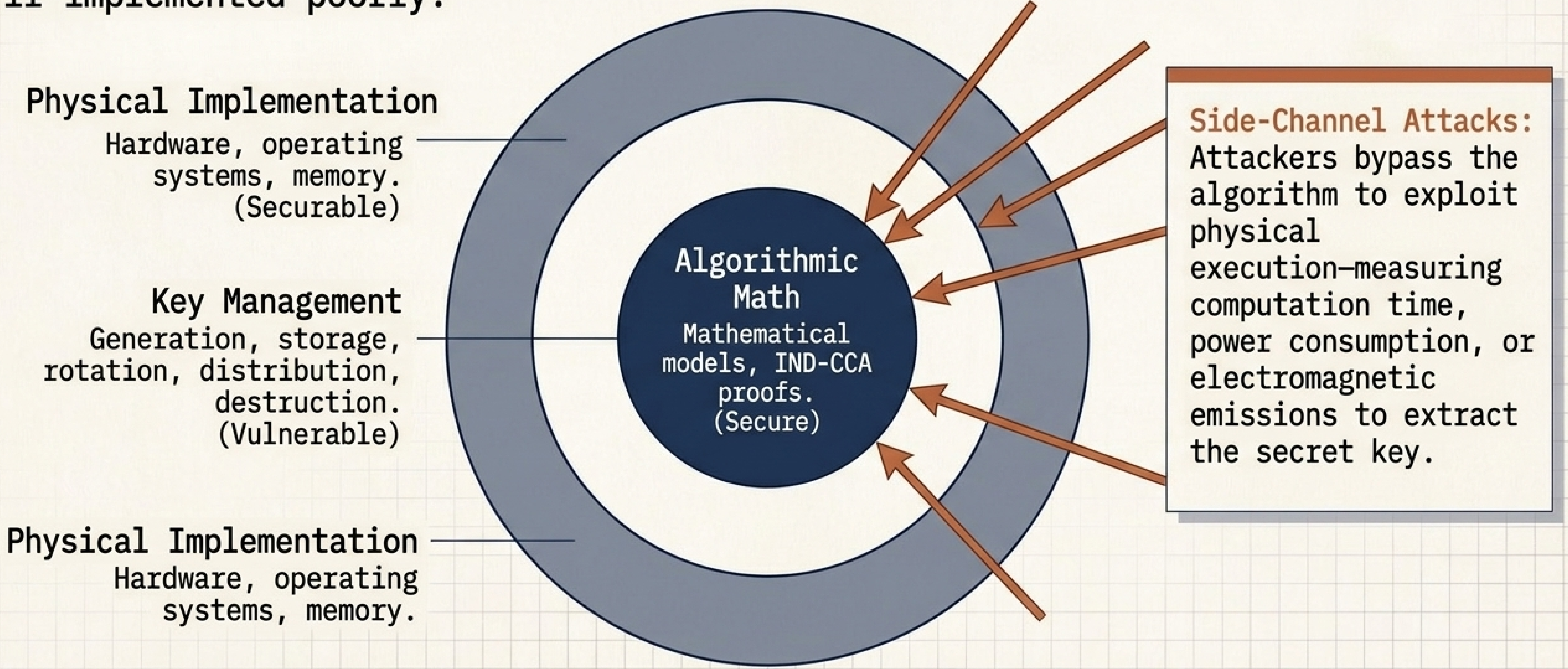
Asymmetric / Public-Key



Uses mathematically related key pairs. Solves key establishment and enables digital signatures without a pre-shared channel. Requires Public-Key Infrastructure (PKI).

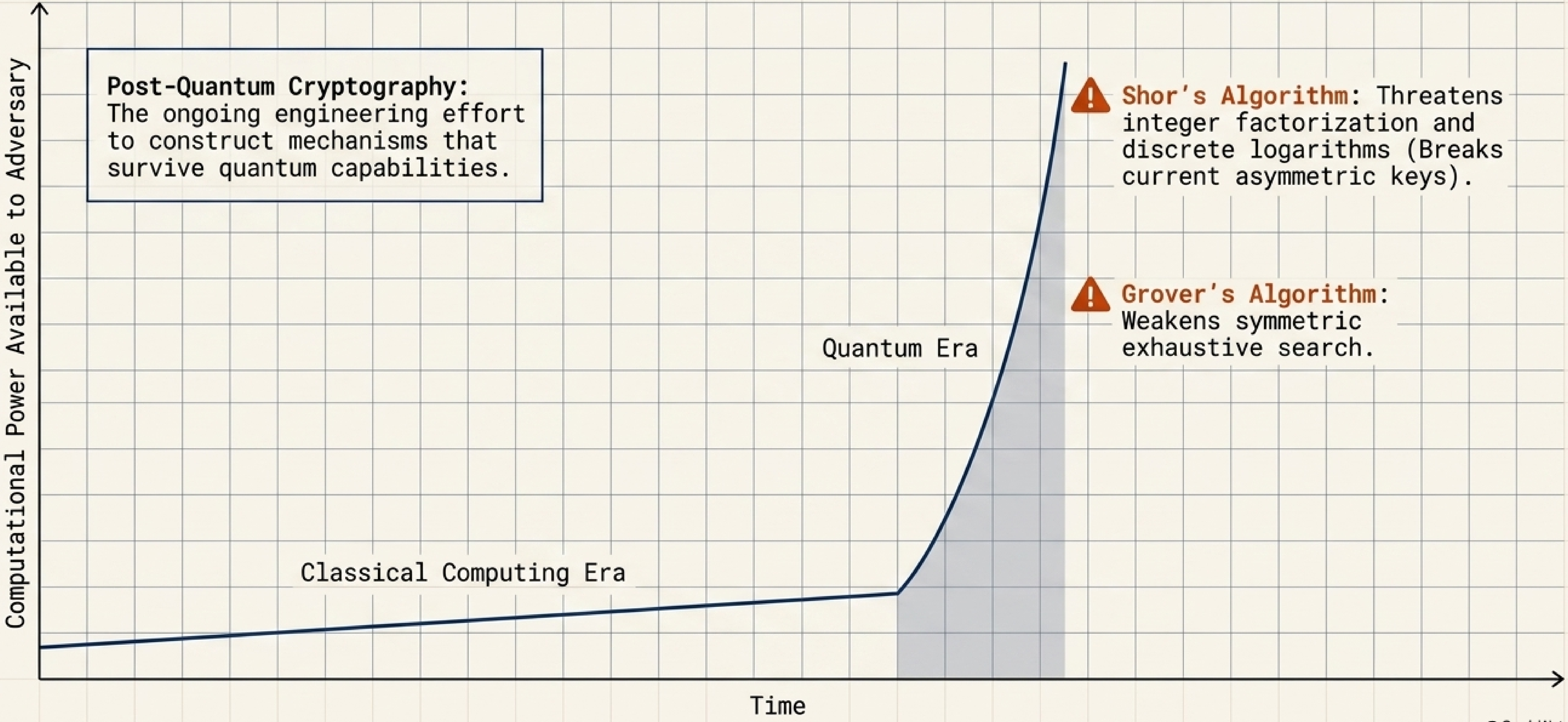
Algorithmic security does not guarantee system security

A mathematically robust primitive can produce a completely insecure system if implemented poorly.



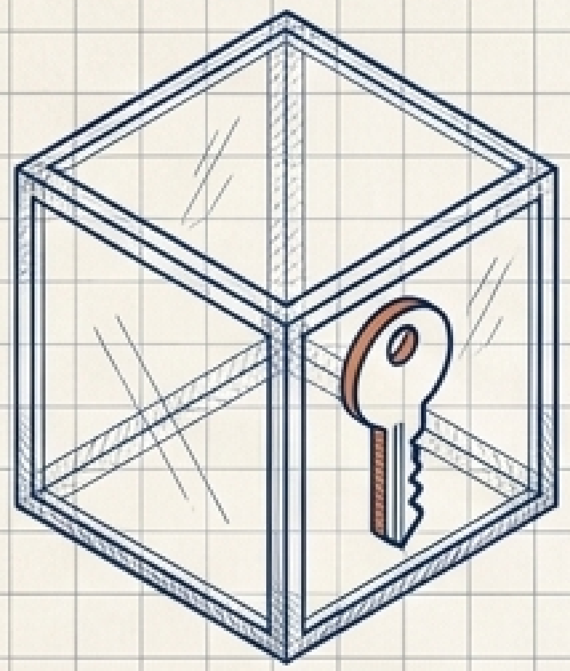
The shifting baseline of computational security

Cryptographic strength is directly tied to the resources required to break it. Security is not static.



Synthesis: The modern cryptographic mindset

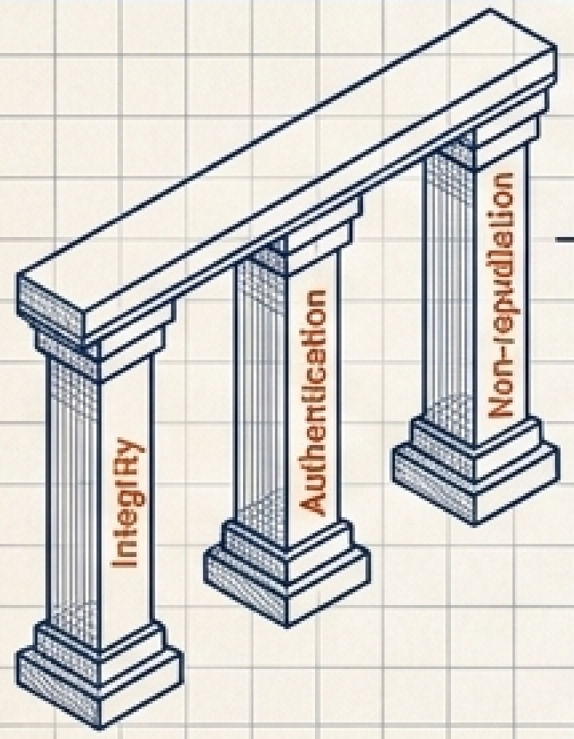
Moving from heuristic obscuration to the rigorous science of trust requires abandoning assumptions.



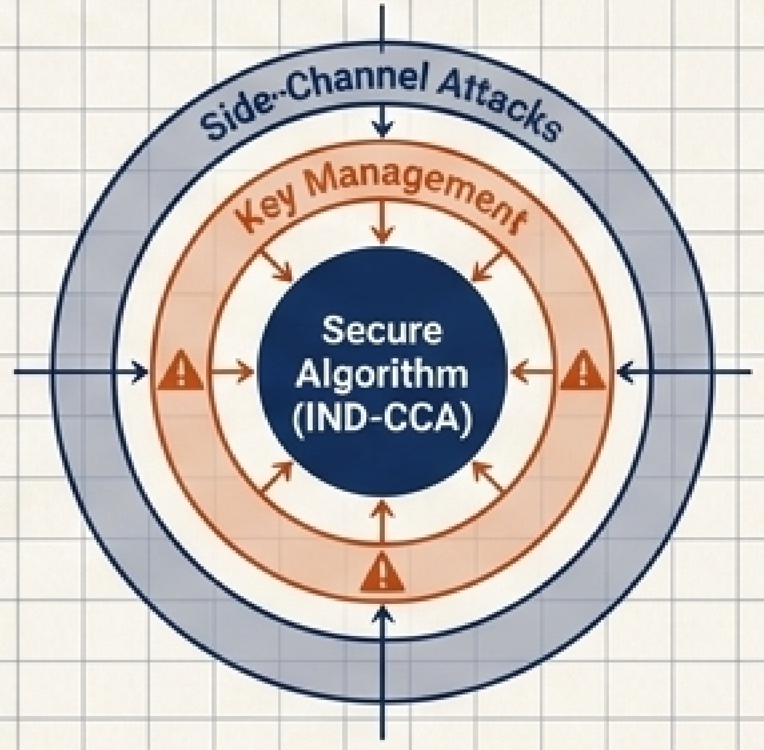
1. Public Constructions: Security relies entirely on secret keys, never on hidden algorithms.

System Component	Attacker Capability	Status
Encryption	CPA	[SECURE]
MAC	CMA	[SECURE]
Key Exchange	CCA	[VULNERABLE]

2. Explicit Adversaries: Systems must be proven secure against actively defined attacker capabilities (e.g., CPA, CCA).



3. Precise Definitions: Protection goes far beyond hiding messages, encompassing integrity, authentication, and non-repudiation.



4. Holistic Implementation: A secure algorithm is useless without robust key management and protection against physical side-channel attacks.

Cryptography is the ever-evolving science of trust.

Modern cryptography does not depend on an algorithm merely looking complicated. It depends on public constructions, properly managed keys, precise definitions, and rigorous mathematical analysis.

The objective is no longer to hide the writing—it is to systematically engineer trust in a hostile environment.